Pranesh U S

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Education

Sri Krishna College of Engineering and Technology — Tamil Nadu, India
Bachelor of Technology - Computer Science and Business Systems; CGPA: 8.21 — *2024 – 2028*

Experience

Software Engineering Intern — Jan 2026 – Mar 2026
Infosys Springboard — *Virtual*

- Secured **MedVault**, a healthcare platform, by integrating **Spring Security, JWT, and RBAC** across **REST APIs** and **MySQL** databases.
- Architected a **refresh-token authentication flow** with blacklist-based logout, reducing unauthorized access vulnerabilities by **90%**.
- Designed **normalized MySQL schemas** for secure patient record retrieval, cutting query latency by **30%** across **10,000+ records**.

Java Development Intern — Jun 2025 – Aug 2025
Pinnacle Lab — *Remote*

- Built **Flight Booking** and **Library Management** systems using advanced **Java OOP** principles, improving code reusability by **40%**.
- Developed **SeedCart**, a **Spring Boot** backend automating dynamic database seeding for **50+ test scenarios**, cutting setup time by **60%**.
- Implemented **layered backend architectures** and optimized **REST API endpoints** for scalable data handling in high-traffic modules.

Projects

VPC-Native AI Gateway – Secure BPO Infrastructure | *Java, Spring Boot, React.js, Redis, Kafka, Presidio* — *2026*

- Engineered a **secure, VPC-native AI gateway** for BPOs so sensitive customer data never leaves the private network when calling LLM APIs.
- Integrated **Microsoft Presidio** for **task-aware PII masking** and **Shannon Entropy-based smart model routing** to pick the right model per request.
- Implemented a **SHA-256 hash-chained audit ledger** with asynchronous **Kafka** streaming to ensure tamper-proof **DPDP Act** compliance.
- Built a real-time **Redis micro-billing ledger** and a **dual-filter security scanner**, reducing AI API costs by **40%** while preventing hallucinated data leaks.

The Debugger Tool – Automated Error Resolution | *Python, React.js, AST Parsing, Regex* — *2026*

- Engineered a **debugging assistant** that parses complex **stack traces** and source code using **Abstract Syntax Tree (AST)** analysis to pinpoint root causes.
- Developed an **error-mapping algorithm** to trace exception propagation across multiple files, reducing manual debugging time by **45%**.
- Used **Regex**-based pattern matching to extract error types, file paths, and line numbers from varied language and framework traces.
- Created an interactive **React.js** UI to visualize **error propagation trees** and apply targeted code patches, speeding up the fix cycle.

AuthForge & SecureBlog – Security Suite | *Java, Spring Boot, React.js, PostgreSQL* — *2025 – 2026*

- Built **AuthForge**, a reusable **Spring Security** authentication framework supporting **JWT, access/refresh tokens**, and custom filters, reducing integration time by **50%**.
- Developed **SecureBlog**, a full-stack platform with **role-based access control**, rich text editing, and real-time search, tested with **500+ concurrent user simulations**.
- Optimized **PostgreSQL** schemas and packaged reusable authentication modules for secure, scalable development with minimal configuration.

Certifications

- **Claude Code in Action** (2026) | **Intro to LLMs** – Google Cloud (May 2025) | **Python for Data Science** – IBM (Aug 2025)
- **Software Engineering, Intro to AI, DB & SQL** – Infosys (2026) | **Networking Basics** – Cisco (Jan 2025)

Technical Skills

Programming Languages: Java, Python, JavaScript, SQL, HTML, CSS
Databases: MySQL, PostgreSQL, Firebase, Redis
Frameworks: Spring Boot, Spring Security, React.js, FastAPI, Tailwind CSS
Tools/Technologies: AWS (EC2, IAM, S3), Docker, Apache Kafka, Git, GitHub, Postman, REST APIs, JWT, OAuth2
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming (OOP), Microservices, VPC Architecture